Comparison of Various Closed Loop Controllers with a Motor/Flywheel System

Addeo, Bryce; Rosenblum, Rex
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***Abstract*—This lab investigated the performance of several feedback control strategies—bang-bang, proportional, Proportional Integral Derivative (PID), and full-state-space—applied to a motor/flywheel system. Each method was evaluated to compare their effects on tracking accuracy, response speed, and control effort. Each controller was tested using 0.25 Hz square and sine wave inputs. The bang-bang controller produced the fastest response, with rise time as low as 0.10 seconds, but exhibited the largest Root Mean Sum (RMS) errors up to 130°, greater motor effort, and overshoots of 25% due to its on-off switching nature. The proportional controller achieved improved accuracy, reducing RMS error to 25° at K_p = 5.0, though with slower rise times around 0.74 seconds. The tuned PID controller with gains of K_p = 8, K_i = 35, and K_d = 0.35, and the full-state-space controller with gains of K_p = 18.7165, K_i = 0, and K_d = 0.5679, provided the most balanced performance, achieving quick settling with minimal steady-state error and smoother control signals.**

***Index Terms*—Bang-Bang, Motor Controller, PID Tuning, State Space Model**

I. Introduction

THE main objective of this lab was to design, implement and compare several feedback control strategies—bang-bang, proportional, PID, and full-state-space—on the motor/flywheel system from Lab 1. Each strategy controlled the rotation of the flywheel by minimizing the error for both square and sine wave inputs. Second order parameters such as rise time, overshoot, and settling time, as well as command effort, were compared to evaluate how the different control strategies affected the overall performance of the system in terms of dynamic response, tracking accuracy and energy efficiency.

The bang-bang controller method subtracts the desired input from the current output to get an error. Depending on if the error is positive, negative, or zero, the controller outputs a new signal to drive the input closer to the desired output. This input is essentially 100% positive or 100% negative hence the name bang-bang. This method can often lead to oscillation around an error of zero known as chatter [1] due to the full on or off method of control so a deadband is usually implemented. A deadband is a set range around zero error where the controller will treat the error as zero instead of trying to get even closer. This control law can be seen in (1) where *u(t)* is the output command, *c* is the level of command effort, *d* is the deadband, and *e(t)* is the error, calculated as the desired rotation, $\theta_{desired}(t)$, minus the actual rotation, $\theta_{actual}(t)$, as shown in (2) below.

$$u(t) = \begin{cases} +c & e(t) \geq d \\ 0 & -d < e(t) < d \\ -c & e(t) \leq -d \end{cases} \tag{1}$$

$$e(t) = \theta_{desired}(t) - \theta_{actual}(t) \tag{2}$$

PID control utilizes three different methods of error manipulation then multiplies them by a set gain and sums them up to create a new output. The components include P (Proportional), I (Integral), and D (Derivative). Proportional simply multiplies the current error by a set gain, integral takes the integral of a set number of recent errors to help identify steady state errors then multiplies it by a gain, and finally, derivative approximates the future error based on the current and recent error then multiplies it by a gain. In this lab, both full PID control was used as well as only proportional control keeping I and D equal to zero. The control law for PID shown in (2) displays how each of the three approaches are summed to get the overall response. Below, *u(t)* is the output command, K_p is the proportional gain, K_i is the integral gain, K_d is the derivative gain, *e(t)* is the error, and *T* is the time over which the integral is taken.

$$u(t) = K_p e(t) + K_i \int_{t-T}^{t} e(t)dt + K_d \dot{e}(t) \tag{2}$$

The final control strategy used in this lab was the full-state-space method which involved creating a state space model of the system and determining the ideal gains based on the desired percent overshoot and settling time using pole placement or Ackermann's method. The general equation of the state-space model is shown in (3) below where $\underline{\dot{x}}$ is the derivative of the state vector that describes how the system moves over time, *A* is the system matrix, $\underline{x}$ is the state vector comprised of the error and its derivative for this experiment, *B* is the input matrix including the systems gain and time constant, and $\underline{u}$ is the input vector with the external controls.

$$\underline{\dot{x}} = A\underline{x} + B\underline{u} \tag{3}$$

II. Procedure

The required materials for this lab were a DAQ, laptop capable of running LabVIEW, a USB-C extension cord, and the motor/flywheel system from Lab 1 comprised of a horizontal flywheel, DC motor, gear set, bearings and a rotational encoder [2]. The procedure for this lab included adding controllers to and modifying the LabVIEW VI, recording data using different feedback strategies, and calculating parameters from the data to evaluate controller performance.

LabVIEW VI Setup

The LabVIEW VI from Lab 1 was modified to implement closed-loop feedback control for the motor/flywheel system. First off, the command signal indicator from last lab was renamed as desired rotation, and then an error indicator was created as the difference between the desired angle and the measured angle from the encoder. This error was then routed as an input to two different control methods—bang-bang and PID. The bang-bang method was added by implementing (1) into the VI with *c* equal to 1023, representing the max power that could be supplied to the motor. The PID control was added by implementing (3) with proper connections to terminals on a Math Tool block. The number of points to integrate and differentiate over were set equal to 100 and 5, respectively. A switch was added to select between these two control methods and an indicator for command signal, *u(t)*, was added as the output which was sent to the motor driver. The new command signal and error were made into local variables (changed to read) and connected to terminals on the data logging block to ensure their values were saved for analysis.

Bang-Bang Control Testing

The first feedback method tested in this experiment was the bang-bang control. On the front panel of the VI, the Turn On Command was set to zero, the frequency was set to 0.25 Hertz, the amplitude was set to 117° and the counts/rev control was set to 2196. The control and input switches were turned to bang-bang and square wave, respectively. First, the motor effort control was set to 0.50 (50% motor output) and the deadband control was set to 0°. The VI was initiated and the response was given enough time to reach steady state. Once steady state was achieved, the save data button was pressed and four cycles of data were taken before stopping the program. The data was saved with a .xls filename extension for later analysis. This same process was repeated for deadband values of 5° and 10° for 50% motor output. Then, the motor effort control was changed to 1.0 (100% motor output) and the same process of saving data was done for the three deadband values. Once all the square wave data was taken, the input was switched to sine wave, and the same process was repeated for the six test cases.

Proportional Controller Testing

The control method switch was turned to PID, but only the proportional controller, K_p, would be used for the following test cases. All values previously set on the VI were left the same, and the square wave input was once again selected. The VI was initiated and the K_p control was set to a value of 0.75 and time was provided for the response to reach steady state. Four cycles of data were then saved using the same steps described for bang-bang testing. This process was repeated for K_p values of 2.0 and 5.0, and then the input was switched to sine wave and all three test cases were done again.

PID Tuning Method

The next test case involved manually tuning a PID controller by systematically adjusting values for K_p, K_i, and K_d. All values previously set on the VI remained the same, but the input was switched back to square wave and all PID constants were set to zero. The K_p control constant was first incrementally increased until oscillations occurred and was then set to half that value. Next, the K_i control constant was increased until the steady state error was minimized. Finally, the K_d control constant was slowly increased until the response looked acceptable in terms of rise time, overshoot and steady state error. Final adjustments were made as necessary to combat the changes caused by introducing new control constants. This same control was then tested for a sine wave input and four cycles of data were saved.

Full-State-Space Method

The final control method used in this lab was the full-state-space model that involved solving multiple equations and using results from Lab 1 to develop a proportional and differential controller. First, the damping ratio, ζ, was solved using (4) where the percent overshoot, *P.O.*, was equal to 5% (0.05). Next, the natural frequency, ω_n, was solved using (5) with a settling time, $T_{s2\%}$, equal to 0.3 seconds. Finally, the damped natural frequency, ω_d, was solved by using the results from (4) and (5) as shown in (6) below.

$$\zeta = \sqrt{[(\ln\left(\frac{P.O.}{100}\right))^2] * [\pi^2 + \left(\ln\left(\frac{P.O.}{100}\right)\right)^2]^{-1}} \tag{4}$$

$$\omega_n = 4 * (\zeta * T_{s2\%})^{-1} \tag{5}$$

$$\omega_d = \omega_n\sqrt{1-\xi^2} \tag{6}$$

With the above parameters solved for, two poles, s_1 and s_2, of the motor/flywheel system were found using (7) below where *j* is the imaginary unit, and then these two poles were assembled into a 1 x 2 pole matrix, *P,* as shown in (8).

$$s_1 = -\xi\omega_n + j\omega_d, \qquad s_2 = -\xi\omega_n - j\omega_d \tag{7}$$

$$P = [s_1 \quad s_2] \tag{8}$$

Next, *A* and *B* matrices were found using the time constant, τ, and the DC gain, K_{DC}, from Lab 1 equal to 0.066 seconds and 1.3375 rad/Vs, respectively, as shown in (9) and (10) below.

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -1/\tau \end{bmatrix} \tag{9}$$

$$B = [0 \quad K_{DC}/\tau] \tag{10}$$

Once these matrices were found, MATLAB was used to calculate K_p and K_d with the command acker[A,B,P] after properly defining the matrices. The result of this command was a 1 x 2 matrix with the first and second terms equal to K_p and K_d, respectively. These values were then added to the corresponding controls in LabVIEW with $K_i = 0$. Four cycles of data were taken and saved with this method for both square and sine wave inputs in the same fashion as the previous test cases.

Data Analysis

Each of the 22 data sets had to be analyzed by solving parameters to evaluate controller performance. First, for every data set, a graph was made on Excel that plotted the platen rotation angle (°) and the command (bits) on the vertical axis against elapsed time in seconds on the horizontal axis. The best singular cycle was chosen, and all remaining data was deleted. The graph axes were then adjusted to best display the single cycle. Next, error and command information was extracted using Excel formulas. The mean and standard deviation were found using the respective Excel functions and inputting the absolute value of all the error and command data values. The root mean square, *RMS*, of the error and command values were then found using (11) below where N is the total data points and x_i is each individual error or command value.

$$RMS = \sqrt{(N^{-1})\Sigma(x_i)^2} \tag{11}$$

Finally, for only square wave data, percent overshoot, *P.O.*, rise time, T_r, and settling time, $T_{s2\%}$, were calculated on Excel. Percent overshoot was calculated by subtracting the desired rotation from the maximum rotation value, dividing that difference by the desired rotation value, and multiplying that quotient by 100 to express the answer as a percentage. Rise time was calculated by taking the difference in time between the data points at 10% and 90% of the maximum rotation value. Finally, settling time was solved by measuring the amount of time between the first and last time that the rotation response crossed the 2% upper/lower boundary before settling.

III. Results

A total of 22 test cases were conducted—12 bang-bang, 6 proportional, 2 PID, and 2 state-space—with half using square-wave inputs and half using sine-wave inputs. For each test, the error (Err.) and command signals (Com.) were analyzed by computing their mean, standard deviation, and RMS values. For the square-wave cases, additional second-order parameters—percent overshoot, rise time, and settling time—were determined. Tables I and II present the bang-bang error and command data for square and sine inputs, while Table III lists the corresponding second-order parameters for square wave bang-bang tests. Tables IV and V summarize the same results for the proportional, PID, and state-space controllers, and Table VI shows their second-order characteristics. Negative overshoot values indicate the desired rotation was not reached, while settling times labeled N/A or 0 correspond to responses that never settled or settled immediately, respectively.

TABLE I
SQUARE WAVE BANG-BANG ERROR AND COMMAND METRICS

Gain	Dead-band (°)	Mean Err. (°)	σ Err. (°)	RMS Err. (°)	Mean Com. (bit)	σ Com. (bit)	RMS Com. (bit)
0.5	0	39.37	67.69	129.9	511.5	0	161.6
0.5	5	35.15	65.83	14.16	147.6	231.8	5.72
0.5	10	37.25	65.94	22.72	129.1	222.2	17.17
1.0	0	32.28	55.92	79.53	1023	0	549.4
1.0	5	24.82	43.48	10.50	858.1	376.2	68.69
1.0	10	22.80	50.77	316.1	245.4	436.8	91.62

TABLE II
SINE WAVE BANG-BANG ERROR AND COMMAND METRICS

Gain	Dead-band (°)	Mean Err. (°)	σ Err. (°)	RMS Err. (°)	Mean Com. (bit)	σ Com. (bit)	RMS Com. (bit)
0.5	0	5.80	3.28	8.42	511.5	0	721.0
0.5	5	3.44	2.65	33.90	155.3	235.2	569.9
0.5	10	7.75	3.17	25.98	140.2	228.2	22.89
1.0	0	12.36	6.36	1.52	1023	0	336.5
1.0	5	11.15	5.64	18.47	830.1	400.2	1236
1.0	10	5.26	3.47	94.32	138.2	349.8	320.4

TABLE III
SQUARE WAVE BANG-BANG SECOND ORDER PARAMETERS

Gain	Deadband (°)	P.O. (%)	T_r (s)	$T_{s2\%}$ (s)
0.5	0	8.736	0.210	N/A
0.5	5	7.623	0.205	0.127
0.5	10	3.987	0.197	0
1.0	0	24.831	0.105	N/A
1.0	5	24.358	0.103	N/A
1.0	10	23.748	0.102	0.498

TABLE IV
SQUARE WAVE P, PID, AND FULL-STATE ERROR AND COMMAND METRICS

Type	Gain	Mean Err. (°)	σ Err. (°)	RMS Err. (°)	Mean Com. (bit)	σ Com. (bit)	RMS Com. (bit)
P	0.75	112.6	4.60	98.16	84.45	3.45	73.62
P	2.0	72.41	41.52	36.42	144.8	83.05	72.83
P	5.0	34.40	44.11	25.05	172.0	220.5	125.3
PID	*	17.84	51.81	35.97	173.7	493.3	346.4
FSS	**	20.97	49.66	7.52	388.5	239.6	145.4

TABLE V
SINE WAVE P, PID, AND FULL-STATE ERROR AND COMMAND METRICS

Type	Gain	Mean Err. (°)	σ Err. (°)	RMS Err. (°)	Mean Com. (bit)	σ Com. (bit)	RMS Com. (bit)
P	0.75	78.56	46.36	343.0	58.92	34.77	257.3
P	2.0	61.84	29.78	86.54	123.7	59.57	173.1
P	5.0	32.00	14.24	38.36	160.0	71.20	191.8
PID	*	15.11	5.88	35.13	178.4	70.36	322.0
FSS	**	9.44	3.46	10.45	173.7	64.16	187.0

TABLE VI
SQUARE WAVE P, PID, AND FULL-STATE SECOND ORDER PARAMETERS

Type	Gain	P.O. (%)	T_r (s)	$T_{s2\%}$ (s)
P	0.75	-92.707	0.452	0
P	2.0	-34.980	0.740	0
P	5.0	-14.383	0.273	0
PID	*	-0.647	0.109	0
FSS	**	-3.455	0.121	0

*Gain values are $K_p = 8$, $K_i = 35$, $K_d = 0.35$
** Gain values are $K_p = 18.7165$, $K_i = 0$, $K_d = 0.5679$

IV. Discussion

Restatement of Objectives

The objective of this lab was to implement and compare several feedback control strategies—bang-bang, proportional, PID, and full-state-space—to regulate the rotation of the motor/flywheel system from Lab 1. Each controller was tested using square and sine wave inputs to minimize the position error between the desired and measured rotation angles. Performance was evaluated through comparisons of error, rise time, settling time, and control effort to assess how each control approach influenced system response, tracking accuracy, and efficiency.

Comparison of Bang-Bang and Proportional Controllers

The bang-bang and proportional controllers exhibited distinct differences in tracking accuracy, control effort, and system response. The bang-bang controller produced larger RMS errors—up to 130° during square-wave tests—while the proportional controller reduced error to about 25° at higher gains, resulting in smoother and more accurate tracking. Bang-bang control achieved faster rise times (0.10–0.21 s) compared to proportional control (0.27–0.74 s), but with greater instability and positive overshoot caused by its on–off switching that caused the system to drive past the target angle before reversing direction. In contrast, the proportional controller's continuous response and higher damping led to negative overshoot, as the system undershot and gradually approached the target. Three of the six bang-bang tests failed to settle within a 2% upper/lower boundary, while all proportional cases settled immediately within the 2% boundaries. Bang-bang control also required greater effort, with RMS commands exceeding 500 bits versus 70–125 bits for proportional control. Overall, proportional control provided higher precision and efficiency, while bang-bang offered faster yet less stable motion, illustrating the trade-off between speed and stability.

Effort Level Effect on Bang-Bang Control

For each bang-bang test case the motor effort was either set to 0.5 (50% motor output) or 1.0 (100% motor output). Increasing the motor effort level from 50% to 100% produced a noticeably faster but more oscillatory response. Rise time decreased from approximately 0.20 seconds at 50% effort to about 0.10 seconds at 100% effort, indicating that the higher torque allowed the system to reach the desired position more quickly. However, this came at the cost of stability—percent overshoot increased from about 4-8% at 50% to as high as 25% at 100%, and RMS error values rose from roughly 79° to 130°. Additionally, the RMS command effort nearly doubled from around 310 bits to over 550 bits, reflecting the stronger and more abrupt motor activation. Overall, higher effort improved responsiveness but introduced greater oscillations and energy usage, while lower effort produced slower yet smoother control behavior.

Proportional Controller Reaction to Gain

The proportional controller behaved as expected as gain increased during testing. The mean error decreased from 78.56° at $K_p = 0.75$ to 32.00° at $K_p = 5.0$ in the sine-wave tests, reflecting improved responsiveness at higher gains. At low gains, the system was sluggish and unable to reach the desired position in time, resulting in larger errors. The mean command signal also increased with gain, as the controller applied greater effort to achieve faster responses. Additionally, the negative percent overshoot rose from −92.71% at 0.75 gain to −14.38% at 5.0, showing that higher gains reduced undershoot and allowed the response to approach the target more closely. Overall, increasing K_p enhanced tracking accuracy and responsiveness at the expense of higher control effort.

Manually Tuned PID Versus Full-State-Space Methods

The manually tuned PID controller was found to have higher errors and larger required effort than the full-state-space controller in response to both square and sine wave inputs. Comparing the square wave, the RMS of the error with PID control had a value of 35.13° while the FSS had a value of 10.45° showing the FSS gave more accurate positioning with lower average error. Moving onto the RMS of the command, the PID had a value of 346.4 bits indicating a large amount of motor effort required to get the desired output on average. Meanwhile, the FSS had a value of 145.4 bits showing that this method was able to achieve the same desired output while working less. Looking to settling time and percent overshoot, both methods had well refined gain values, resulting in settling time being calculated as zero. For percent overshoot, the PID controller had a value of -0.647% while the FSS had a value of -3.455%. Both values being negative indicated that the system was slightly over damped leading to a slower than perfect rise to the commanded value. However, the manually tuned PID had a value much closer to zero while still being negative showing a better response than the FSS controller.

Limitations and Suggestions for Improvement

The largest limitation observed during this experiment came from the hardware of the motor/flywheel system. The system utilized 3D printed ball bearings which led to a rougher than ideal surface and thus, more friction. Adding to this issue was the fact that the bearing assembly was held together with bolts that did not limit the applied pressure on the bearing. These factors created a system that was difficult to apply the appropriate pressure to and had an imperfect bearing surface. These limitations prevented the system from overcoming internal friction at lower gain values during the proportional controller testing resulting in a high negative percent overshoot or no rotation response at all. In future tests, an off-the-shelf thrust bearing paired with a more controllable mounting system would greatly decrease system friction and allow for more accurate results when testing control methods.

References

[1] "EML 4314C - Fall 2025 - Lab #2 Assignment" University of Florida, 2025. [Online].

[2] "Pendulum v2.3 Assembly Instructions," Out of the Box Electronics. https://www.ootbelec.com/